

Drug Use and Mental Health: Comorbidity between Substance Use and Psychiatric Disorders

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Maria A. Sullivan

Introduction

The co-occurrence of substance use disorders (SUDs) and non-substance use psychiatric disorders is commonly referred to as dual diagnosis. This comorbidity is most common for mood disorders, especially depression. Along with environmental risks, a genetic predisposition is believed to underlie the development of comorbid disorders [46]. Dual diagnosis is of great concern because it is often associated with higher disease severity, poor physical and social functioning, increased rates of psychiatric hospitalization including emergency admissions, self-harm, and suicide [39, 81, 83, 98]. In addition, comorbid drug users have an increased rate of high-risk behaviors and sexually transmitted infections (e.g., HIV), as well as more psychosocial impairments such as unemployment and homelessness, and high rates of violent and criminal behavior [98].

Clinical management of individuals with dual diagnosis can pose unique challenges. In a large US national sample, Krawczyk et al. [50] found that 28% of patients discharged from substance treatment facilities had psychiatric comorbidity and 38% did not complete treatment. Clients with

psychiatric comorbidity had higher odds of not completing either alcohol or substance treatment, as well as an earlier time to attrition, relative to those without comorbidity. Individuals with psychiatric comorbidities seeking or receiving treatment for substance use disorders face particular challenges that affect their ability to complete treatment. And conversely, the presence of a comorbid substance use disorder is associated with a poor response to treatment for many psychiatric disorders. In this chapter, we describe evidence for the prevalence of co-occurring psychiatric disorders and substance use disorders and the effects of this comorbidity on treatment course. We suggest that these findings call for further efforts to integrate treatment for psychiatric comorbidities in substance use treatment settings.

Epidemiology: Prevalence of Psychiatric and Substance Use Disorder Comorbidity

Comorbidity of Substance Use Disorder (SUD) and Anxiety Disorders

Numerous studies with different methods have shown a significant association between anxiety disorders and alcohol use disorders [77]. Recent epidemiological studies have reported lifetime prevalence rates of 23.5% alcohol dependence and 20.4% alcohol misuse in patients with social

M. A. Sullivan (✉)
Department of Psychiatry, Columbia University
College of Physicians and Surgeons, New York State
Psychiatric Institute, New York, NY, USA
e-mail: mas23@cumc.columbia.edu

phobia [27]. Conversely, the lifetime prevalence of social anxiety disorder among patients with alcohol dependence is about 11% [70]. Anxiety symptoms are associated with a worse course of alcohol use disorder, marked by early relapse following detoxification [88].

Using data from the 2001–2002 National Epidemiologic Survey on Alcohol and Related Conditions ($N = 43,093$) and the Alcohol Use Disorder and Associated Disabilities Interview Schedule-DSM-IV Version as the diagnostic instrument, Alegria et al. [2] determined that GAD-SUD (2.04%) constitutes half of the lifetime prevalence of GAD (4.14%) in the US adult population. With respect to lifetime risk factors, individuals with GAD-SUD were more likely than those with GAD-NSUD to have had a vulnerable family environment, a family history of antisocial behavior, and a family history of AUD/SUD. The onset of SUD preceded the onset of GAD among those with GAD-SUD, suggesting that SUD, at least in some cases, may facilitate the initiation of GAD. Furthermore, GAD-SUD is associated with high overall vulnerability for additional psychopathology, particularly in the externalizing spectrum, as well as higher disability. Individuals with GAD-SUD also had significantly higher rates of use of alcohol and drugs to relieve symptoms of anxiety among those with GAD-NSUD; this behavior may provide temporary relief from the anxiety symptoms, but may increase symptoms in the long term. Taken together, these findings suggest a stronger predisposition for psychopathology among individuals with GAD-SUD that is further exacerbated by the use of substances [2].

A complex comorbidity exists between alcohol use disorder (AUD) and post-traumatic stress disorder. Because of the associations between at-risk drinking, violence, and accidents, individuals with AUD are frequently exposed to extreme stress and the subsequent development of PTSD [13, 58]. The externalizing disorders common in AUD (e.g., attention-deficit/hyperactivity disorder, conduct disorder, and antisocial personality disorder) are also associated with impulsive and risk-taking behaviors leading to trauma [16].

Post-traumatic stress disorder (PTSD) also conveys an increased risk for the development of substance use disorders (SUD). In patients diagnosed with PTSD, the prevalence of comorbid substance use ranges from 19% to 35% and comorbid alcohol abuse ranges from 36% to 52% [82, 84]. Among patients with PTSD, opioid use disorder is a less common but increasing comorbidity. In a large patient population of veterans who initiated PTSD treatment in the Department of Veterans Affairs (VA) between 2004 and 2013 ($N = 731,520$), comorbid opioid use disorder diagnoses increased from 2.5% in 2004 to 3.4% in 2013. Patients with comorbid opioid use disorder used more health services and had more comorbidities than other patients with PTSD [91].

Emotional processing theory posits that PTSD symptoms are maintained, in part, by negative trauma-related cognitions about the world as entirely dangerous and the self as entirely incompetent [25]. Recovery from PTSD therefore involves modifying these cognitions; this is the mechanism by which negative cognitions mediate change in PTSD symptoms during exposure therapy [63]. PTSD often precedes SUD, and a recent analysis revealed that frequency and intensity of negative alterations in cognition and mood predicted alcohol use disorder (AUD), but re-experiencing symptoms, hyperarousal, and avoidance did not [11]. These findings suggest that those who experience negative alterations in cognition and mood may be at increased risk of developing AUD, and that we may be able to predict which individuals in clinical settings will be strong candidates for new combined PTSD/SUD treatments. For instance, prolonged exposure therapy and oral naltrexone for reducing alcohol use among those with comorbid PTSD/AD exert a combined effect through reduction in both PTSD symptoms and craving (McLean et al. 2015).

Comorbid PTSD/SUD is associated with a more complex and costly clinical course when compared with either disorder alone, including increased chronic physical health problems, poorer social functioning, higher rates of suicide attempts, more legal problems, increased risk of

violence, worse treatment adherence, and less improvement during treatment [57]. In the treatment of comorbid PTSD/SUD, pharmacotherapy should not replace trauma-specific psychotherapy [77], and psychosocial interventions are the preferred treatment model.

Comorbidity of SUD and Mood Disorders

In the United States, the lifetime prevalence of illicit drug abuse is estimated at 19.4% in mood disorders [47] and 24% in major depressive disorder [104]. The presence of drug abuse has been shown to increase the risk for depression five-fold [81, 83]. Using the 2001–2 National Epidemiologic Survey on Alcohol and Related Conditions (NESARC) as a representative sample of the US adult population ($N = 43,093$), Blanco et al. [10] found that the lifetime prevalence of major depressive disorder without SUD (MDD-NSUD) was 7.41%, whereas that of MDD-SUD was 5.82% and the prevalence of substance-induced mood disorder was quite low, at 0.26%. The researchers noted that a family history of psychopathology and most childhood and adulthood risk factors were more common in MDD-SUD than in MDD-NSUD. Similarly, Sher et al. [90] reported that individuals with MDD and alcohol use disorder (AUD) were more likely to have a family history of AUD and abuse during childhood than those with MDD and no AUD comorbidity. The results by Blanco et al. [10] suggest that SUD is a marker for greater severity in individuals with MDD; MDD-SUD is characterized by earlier age at MDD onset, higher number of MDD criteria met, higher number of depressive episodes, and higher rates of psychiatric comorbidity, compared to MDD-NSUD. Further, some risk factors – such as a family history of psychopathology -- may not be disorder-specific but, instead, shared between MDD and SUD [60].

Compared to other psychiatric disorders, the rates of comorbidity with alcohol use disorder are highest in bipolar disorder. In the Epidemiological Catchment Area (ECA) Study

[81, 83], frequencies of 46.2 and 39.2%, respectively, for bipolar I and II were reported. Recent epidemiologic studies (NESARC; [8]) found frequencies of 54.6% and 51.8% for bipolar I and bipolar II disorders. Among bipolar patients, male gender, history of higher number of manic episodes, and previous history of suicidality are associated with higher susceptibility to substance use disorder (SUD). Thus, in individuals with bipolar disorder at increased risk of drug abuse, more intensive therapeutic interventions should be considered to prevent development of SUD [61]. Further, bipolar disorder is associated with an increased risk of behavioral addictions; pathological gambling and kleptomania are the most prevalent conditions, followed by compulsive buying, compulsive sexual behavior, and internet addiction [103].

A recent study examined co-occurrence of tobacco use, substance use, and mental health problems, and its moderation by gender, among 32,202 US adults from Wave 1 (2013–2014) of the nationally representative longitudinal Population Assessment of Tobacco and Health (PATH) Study. In this nationally representative sample of US adults, Conway et al. (2017) demonstrated that female tobacco users are at increased risk for substance use and mental health problems.

There has also been recent research examining depressive traits that appear to be prevalent in opioid use disorder. In a large ($N = 1195$) study of Italian heroin addicts, Maremmani et al. [59] identified five personality dimensions: “worthlessness and being trapped,” “somatic symptoms,” “sensitivity-psychoticism,” “panic-anxiety,” and “violence-suicide.” These clusters of traits – which were independent of demographic or clinical characteristics, as well as other substances used -- discriminated patients affected by substance use disorders from those affected by non-substance psychiatric disorders. The researchers concluded that these findings suggest that the SUDs result in a trait-dependent, rather than state-dependent, psychopathology. Similarly, there is evidence that certain personality disorders (i.e., antisocial and schizotypal) are particularly linked with chronicity in

substance use disorders [7]. Even though the types of personality disorders seen in individuals with drug and alcohol use disorder are similar, the prevalence of any personality disorder is higher among patients with drug use disorder than alcohol use disorder [73].

Comorbidity of SUD and Schizophrenia

The schizophrenic psychoses have high comorbidity with both alcohol and substance use disorders. The lifetime prevalence of alcohol-related disorders with psychosis rose from 29% in the 1990s to 51% in 2019 [76]. Individuals with schizophrenia are also more likely than those without the disorder to smoke cigarettes, heavily use alcohol, heavily use cannabis, and use recreational drugs; these factors contribute to the premature mortality and increased disability observed in patients with schizophrenia [35]. Emerging data have revealed that individuals who smoke tobacco also have a two-fold increased risk of onset of schizophrenia spectrum disorders [89]. It appears that the smoking of tobacco may play a role in early information processing deficits observed in schizophrenia [79].

A recent meta-analysis of SUDs in epidemiological and treatment-seeking patients diagnosed with schizophrenia or first episode psychosis [41] found that the prevalence of any substance use disorder (SUD) was 41.7%, followed by illicit drugs (27.5%), cannabis (26.2%), alcohol (24.3%), and stimulant use (7.3%). Patients with SUD had an earlier age of onset of schizophrenia. A meta-regression showed that prevalence increased over time for illicit drugs but not for other substances, including alcohol. Further, a meta-analysis revealed that SUD in schizophrenia is highly prevalent, and rates have not changed over time. This finding indicates substance use disorders are difficult to treat in this patient population and more studies are needed to help develop better prevention, detection, and treatment of SUDs in persons with schizophrenia and comorbid disorders.

In an analysis of studies including individuals with psychosis who had a history of substance use, misuse, and dependency related to alcohol, cannabis, and cocaine, Donoghue et al. [22] analyzed how varying levels of use of each particular substance affected cognitive functioning. In contrast to what was expected, they found that individuals who used certain drugs performed better on some cognitive tests than those who did not use drugs. Specifically, cocaine users had better psychomotor processing speed and attention than non-users, but had deficits in memory and verbal ability. Individuals with psychosis who used cannabis had better overall functioning than those who did not use cannabis. The authors recommended caution in interpreting these findings, as individuals able to seek out and acquire drugs, despite the limitations of their psychosis, may be higher functioning and have more neurological capacity than their non-using peers. They noted that longitudinal studies are needed to enable an in-depth assessment of the extent of impairment resulting from long-term substance misuse.

Four theories have been advanced to explain the pathogenesis of the comorbidity between schizophrenia and SUD: (1) the self-medication hypothesis suggests that the use of substances ameliorates some symptoms of schizophrenia [48, 92], (2) schizophrenia and SUD share a common genetic mechanism involving dysregulation in the dopamine-regulated reward circuitry [14, 31, 85]; (3) diathesis-neural stress model: genetic predisposition, combined with the environmental stressor of chronic substance use, results in the onset of schizophrenia [26, 105]; and (4) impaired social and occupational function, together with a limited social environment, together increase the risk of substance use [75]. Co-occurring SUD is present in half of patients with schizophrenia, which is a 3–four-fold higher prevalence than in the general population [35]. Comorbid substance use disorder has a negative impact on clinical outcomes in patients with schizophrenia, including more severe symptoms, less treatment compliance, increased hospitalizations and medical comorbidities [18, 72, 94].

Comorbidity of SUD and Attention-Deficit/Hyperactivity Disorder

Attention-deficit/hyperactivity disorder (ADHD) is associated with an increased risk of developing SUD later in life [15]. Using the largest available meta-analyses of genome-wide association studies (GWAS) of ADHD ($n = 53,293$) and lifetime cannabis use ($n = 32,330$) to examine their causal relationship, Soler Artigas et al. [93] found that the data supported that ADHD is causal for lifetime cannabis use, with an odds ratio of 7.9 for cannabis use in individuals with ADHD in comparison to individuals without ADHD (95% CI (3.72, 15.51), $P = 5.88 \times 10^{-5}$). These results highlight the clinical importance of assessing for substance misuse in the context of clinical interventions for ADHD.

Patients with ADHD-SUD comorbidity typically have an early onset of substance use disorder and a faster transition from less severe to more serious SUD [24]. In an analysis of Waves 1 and 2 of the NESARC data, [4] found that each additional ADHD symptom was generally associated with a proportional increase in odds of substance dependence. Individuals with persistent ADHD, compared to those with remittent ADHD or healthy controls, are at significantly higher risk of developing an SUD. ADHD persisters have also been found to have higher prevalence rates of nicotine dependence (24.2%) than ADHD remitters (16.7%) and healthy controls (4.3%) [42]. With respect to the impact of ADHD on the course of specific substance use disorder, Estevez-Lamorte et al. [23] analyzed a large ($N = 4975$) sample of Swiss men and found that men with ADHD were more likely to exhibit persistent and risky alcohol and nicotine use, and to mature out of risky cannabis use. Further, they identified that early age of alcohol initiation distinguished between persistence and maturing out of AUD, while the course of nicotine use disorder and cannabis use disorder was related to ADHD symptoms and SUD severity at baseline. These findings suggest that among those with ADHD, substance-specific prevention strategies, particularly if implemented before early adulthood, may

represent a critical intervention for reducing the persistence of substance use disorders [23].

ADHD-SUD comorbidity is highly prevalent in addiction treatment settings; in a meta-analysis of 29 studies, 23.1% of patients with SUD also had adult ADHD [102]. Despite seeking treatment more frequently than those without SUD, adults with ADHD-SUD comorbidity have more difficulty remaining abstinent [53] and report a reduced quality of life with more professional, social, and personal problems [49].

Racial and Ethnic Disparities in Dual Diagnosis

In a study drawing on data from two waves of the NESARC, Szaflarski et al. [95] documented significant associations between nativity, race-ethnicity, and dual diagnosis in the US adult population. Immigrants and some racial-ethnic groups (e.g., people of African and Mexican descents) were less likely than US-natives and Europeans to have a dual diagnosis versus having no SUD or depressive/anxiety disorder alone. Conversely, immigrants and minorities were more likely than their native and European counterparts to have a depressive/anxiety disorder alone rather than a dual diagnosis. There was no nativity-based difference in having a SUD alone versus dual diagnosis, although people of African and Mexican origins were more likely to have this condition, compared with Europeans.

Within a US sample of African American and Caribbean Black men and women, significant challenges in accessing and receiving substance disorder treatment have been identified [80]. Black Americans delay seeking treatment, have shorter treatment durations, and experience significant social and health-related consequences because of their substance use, compared to White Americans [12, 69]. For many Black adults, these consequences include poverty, unemployment, lack of a high school diploma or GED, health ailments, and drug-related legal problems [29, 52]. Redmond et al. [80] found that, among those who sought help, African

Americans most commonly sought help from “other health professionals” (81%), and from self-help groups or other informal sources of care (75.9%). Caribbean Blacks also sought help more often from “other health professionals” (86.7%), followed by mental health professionals (62.2%). African American (32.3%) and Caribbean Black (33.3%) respondents were least likely to seek help from a psychiatrist. These findings suggest that individuals with substance and psychiatric comorbidity may be especially unlikely to receive effective care. Preventive psychoeducation on the value of treatment is needed, as many African American and Caribbean Black individuals may not be aware that avoiding or delaying treatment is associated with a lengthier duration of substance disorder problems, which can have associated health and social consequences [80].

Neurobiology of Comorbid Substance Use Disorders and Psychiatric Disorders

Preclinical work has investigated whether depression-like states increase vulnerability to drug-taking behaviors in animals. It has long been recognized that psychosocial stressors produce depression-like phenotypes in animals. But recent research has demonstrated that post-weaning isolation stress and repeated social defeat generally increases initial drug intake and makes animals more sensitive to the rewarding effects of drugs of abuse, as shown by increased drug self-administration. This increased drug acquisition following post-weaning social isolation may be mediated through changes in protein expression that support potentiated dopamine release in the nucleus accumbens. It is of note that these chronic psychosocial stress paradigms both consistently produce depression-like behaviors in animals and also exert an impact on drug-taking behaviors [64].

The genetic underpinnings for the high comorbidity between psychiatric disorders and substance use disorders have not been fully elucidated. Common genetic factors influencing the co-occurrence of anxiety disorders and SUD,

as well as mood disorders and SUD, have been sought through genome-wide association studies (GWAS). Genome-wide linkage scans have revealed significant quantitative trait loci for drug dependence (14q13.2-q21.2, LOD = 3.322) and a broad anxiety phenotype (12q24.32-q24.33, LOD = 2.918). Significant positive genetic correlations were observed between anxiety and each of three addiction subtypes: lifetime history of alcohol dependence, drug dependence, or chronic smoking ($p_g = 0.550\text{--}0.655$) [40]. A linkage signal for anxiety recently identified on 12q24 spans the location of TMEM132D, an emerging gene of interest from previous genome-wide association studies of anxiety traits [40]. Likewise, in analyses adjusted for MDD status in three alcohol dependence GWAS data sets, Andersen et al. [5] observed significant evidence for an association between the MDD polygenic risk score and alcohol dependence (best $P = 0.007$).

In a genome-wide association study of genetic susceptibility to substance use disorders (SUDs) and other psychiatric disorders, significant associations with SUDs were detected for schizophrenia, using polygenic scores [34]. This finding indicates that SUDs share genetic susceptibility with SCZ to a greater extent than with other psychiatric disorders, including externalizing disorders such as attention-deficit/hyperactivity disorder. These researchers noted that women have a lower probability to develop substance abuse/dependence than men at similar polygenic scores, probably because of a higher social pressure against excessive drug use in women. A genetic risk for schizophrenia has also been linked to prospective cannabis use patterns during adolescence [37]. There are indications that increased risk for substance and nicotine use, found for certain dopaminergic polymorphisms (e.g., DRD4-7 for frequent cannabis use or DRD2 A1-allele for alcohol use), is under the influence of environmental factors such as low parental monitoring or childhood trauma [32]. In addition, life-course effects on alcohol consumption have been reported for dopaminergic genes [33].

Diagnostic Considerations and Therapeutic Approaches

There are two separate types of substance disorders described in DSM-5. There are those that are conditions of use (substance use disorders) and those that are induced by the misuse of substances (e.g., psychosis, withdrawal, anxiety, and dysfunction). Substance-induced mood disorders represent a separate diagnostic. According to DSM-5, independent depression that develops outside the context of drug use cannot be entirely causally attributed to substance use and persists even when abstinence is sustained. By contrast, substance-induced depression, while warranting clinical attention, occurs only within the context of substance use and is expected to resolve with abstinence [3]. When assessing for the presence of psychiatric disorders comorbid with substance use disorders, it is essential to take a careful history and apply DSM-5 criteria for primary vs. substance-induced mood disorder, anxiety disorder, or psychotic disorder. A recent investigation has found that 10% of individuals presenting for substance abuse treatment had possible schizophrenia or psychotic illness; these results support the need for routine screening for psychiatric issues, including schizophrenia/psychosis-like symptoms, in adults entering substance use disorder treatment settings [100]. Although the distinction between substance and non-substance (psychiatric) disorders has important prognostic and therapeutic implications, in clinical practice it can often be difficult to differentiate between these two conditions, as they rely upon chronological and symptom severity criteria that patients may be unable to provide with precision [19]. Overall, DSM categories are broadly defined, and their application relies considerably on clinical judgment.

In a review of the evidence of diagnostic accuracy for screening and diagnostic tests to be used in populations with SUD and severe mental illness (SMI) conditions, Larun et al. [51] identify two screening tests for SUD in patients with SMI: (1) the CAGE (Chemical Use, Abuse, and Dependence Scale) to identify alcohol use disorder, both current and lifetime; and (2) the AUDIT

(Alcohol Use Disorders Identification Test). The authors noted one screening tests for SMI in patients with SUD which has the strongest evidence base: the PDSQ (Psychiatric Diagnostic Screening Questionnaire). Among the available validated diagnostic tests, Larun et al. [51] determined that the PRISM (Psychiatric Research Interview for Substance and Mental Disorders) showed good concordance (kappa 0.63–0.90) when compared to a reference standard (e.g., clinical interview or Structured Clinical Interview for DSM-5, SCID).

It is recommended that co-occurring independent depression be addressed immediately with antidepressants or psychotherapy, while substance-induced depression is best managed by careful observation of its course while focusing primarily on the treatment of the substance use disorder (Nunes et al. 2009). Two meta-analyses have validated the efficacy of antidepressant medication for comorbid depression and alcohol use disorder, with studies showing the strongest effects when patients are required to be abstinent before depression was diagnosed [66]. With respect to other substance use disorders comorbid with depression, Dakwar et al. [19] found that cannabis dependence was highly associated with independent depression ($p < 0.001$), while cocaine dependence was highly associated with substance-induced depression ($p < 0.05$) in a sample ($N = 242$) of cocaine, cannabis, and/or opioid-dependent, treatment-seeking individuals. Independent depression was found to be significantly associated with female gender, higher Hamilton-D score, and post-traumatic stress disorder (PTSD). The finding of a strong association between cannabis dependence and independent depression is consistent with research implicating the endo-cannabinoid system in mood regulation [38]. Future studies should seek to further elucidate the effect of depression category on the prognosis of SUD, its illness course, and response to treatment. Trials of psychotherapeutic or behavioral treatments may enable the development of treatment algorithms that integrate the skill sets of mental health and substance abuse clinicians [65].

Attention-deficit hyperactivity disorder (ADHD) is also highly prevalent in substance use disorder (SUD). In a group of treatment-seeking high-dose benzodiazepine-dependent patients ($N = 167$), 31.7% of the sample screened positive for adult ADHD. Patients who were ADHD-positive showed a significantly higher prevalence of poly-drug abuse than did those who were ADHD-negative [96]. Guidelines recommend that when ADHD coexists with other psychopathologies in adults, the most impairing condition should generally be treated first [43]. Although the diagnostic assessment of ADHD is usually postponed until after a period of abstinence, this practice delays timely treatment. In a recent study of treatment-seeking adult SUD patients with a comorbid diagnosis of adult ADHD ($N = 127$), the authors found that in 95.3% of SUD patients with ADHD, the diagnosis of ADHD remained stable during abstinence, although the subtype of ADHD was less stable between assessments. These findings suggest that SUD patients can reliably be evaluated for ADHD during active substance use [101].

Co-occurring psychiatric syndromes are likely to be important in the search for a nosology grounded in the pathophysiology of the substance use disorders [68]. The high prevalence of comorbid substance and non-substance use disorders points to a common etiology with shared genetic and neurobiological features. And, from a clinical perspective, diagnostic acumen can lead to early identification of the concurrent conditions, thereby permitting timely interventions in which effective treatment of one disorder can result in an improved prognosis for the other disorder.

Treatment Course: Prognostic Implications

In light of the heightened risk for self-medication in patients with anxiety and mood disorders, clinicians should be particularly vigilant at monitoring for the presence of substance use problems early in the course of psychiatric treatment [7]. The simultaneous treatment of comorbid disorders is commonly referred to as integrated treat-

ment. Integrated treatment of dual disorders often involves an interdisciplinary team, including social workers, psychotherapists, counselors, and case managers [44]. In the case of comorbid conditions for which pharmacologic options are limited (e.g., schizophrenia and cannabis use disorder), the intensity of psychotherapies and behavioral treatments must be increased. One behavioral treatment that has been proven highly effective for the treatment of comorbid substance and psychiatric disorders in an intensive outpatient setting is Contingency Management [45].

Among individuals ($N = 507$) seeking outpatient treatment for substance use disorders, Sanchez et al. [87] found that one fifth (21%; $n = 106$) screened positive for depression. After controlling for anxiety and PTSD symptoms, presence of depressive symptoms remained significantly associated with fewer coping strategies ($P = 0.001$), greater impairment in social adjustment ($P < 0.001$), and poorer health status ($P < 0.001$), but not to days of drug use in the last 90 days ($P = 0.14$). The presence of depressive symptoms was associated with fewer coping strategies and poorer social adjustment. Since coping skills are a significant predictor of addiction outcomes, screening for and enhancing coping among depressed patients may be an important evidence-based intervention to improve global functioning among substance abusers, as an adjunct to usual treatment [87].

A cross-sectional study of individuals seeking SUD treatment ($N = 1276$) found that substance-dependent patients had impaired quality of life, especially in the mental component of the SF-36 (Maigre et al. 2017). Impaired physical quality of life was independently associated with medical condition, age, being female, depressive disorder, and anxiety disorder. Depression disorder, any personality disorder, active consumption last month, attention-deficit hyperactivity disorder, anxiety disorder, and suicide attempt were independently associated with worse mental quality of life. These findings emphasize the significance of dual diagnosis in the impairment of health-related quality of life in substance-dependent patients, particularly with regard to the mental component. In addicted patients with low scores

on SF-36, psychiatric comorbidity should be evaluated and treated in an integrated approach.

Recent findings from a register-based cohort-based study in Denmark of people born since 1955 suggest that having any substance use disorder (SUD) is associated with at least a three-fold increased risk of completed suicide, compared to those having no SUD [71]. Alcohol misuse was associated with an increased risk of completed suicide in all populations with hazard ratios (HR) between 1.99 [95% confidence interval (CI) = 1.44–2.74] and 2.70 (95% CI = 2.40–3.04). Other illicit substances were associated with a two- to three-fold risk increase of completed suicide in all populations except bipolar disorder. However, cannabis use disorder was associated with increased risk of attempted suicide only in people with bipolar disorder (HR = 1.86, 95% CI = 1.15–2.99). Alcohol and other illicit substances each displayed strong associations with attempted suicide, HR ranging from 3.11 (95% CI = 2.95–3.27) to 3.38 (95% CI = 3.24–3.53) and 2.13 (95% CI = 2.03–2.24) to 2.27 (95% CI = 2.12–2.43), respectively. Cannabis was associated with suicide attempts only in people with schizophrenia (HR = 1.11, 95% CI = 1.03–1.19) [71].

Blanco et al. [10] found that individuals with MDD-SUD were less likely to receive pharmacologic treatment for depression than those with MDD-NSUD, despite evidence that antidepressants are efficacious for treatment of depressive symptoms and modestly improve SUD (Nunes et al. 2004, [21]). In particular, the use of lithium has been found to result in reduced craving and improved near-term clinical outcomes in patients with comorbid bipolar and substance use disorders [28, 78]. Nonetheless, the finding by Blanco et al. [10] is consistent with the established clinical approach to delay initiating pharmacotherapy in SIDD and encourage abstinence. In this NESARC sample, individuals with SIDD reported greater use of substances to relieve their depressive symptoms, suggesting that in the absence of antidepressant treatment some individuals may resort to self-medication.

By contrast, individuals with generalized anxiety disorder (GAD)-SUD are as likely as those

with GAD-NSUD to receive medication for anxiety symptoms [2]. Alegria et al. suggest caution in the use of benzodiazepine in individuals with GAD-SUD due to the increased risk for dependence of prescription drugs among individuals with SUD [9]. Thus, antidepressants (e.g., SSRIs), many of which are effective for GAD, may be preferable as first-line treatment for most individuals with GAD-SUD [67].

It is also important to recognize that while all antipsychotics improve the positive symptoms of schizophrenia, second generations appear most effective in reducing cravings in SUDs. The atypical antipsychotic clozapine has been found to be the most effective medication in reducing alcohol and cannabis use [30, 56, 86], while findings on other medications are equivocal [45]. A systematic review of pharmacological approaches to schizophrenia with comorbid SUD found that the efficacy of clozapine for SUD improvement in schizophrenic patients was superior to that of first-generation antipsychotics in polysubstance users. When compared to second-generation antipsychotics, clozapine was superior to risperidone but equal to olanzapine or ziprasidone in polysubstance and cannabis users [6].

The presence of psychiatric comorbidity exerts a negative influence on SUD prognosis and treatment outcome. Krawczyk et al. [50] found that 28% of patients discharged from substance treatment facilities had psychiatric comorbidity, and 38% did not complete treatment. Clients with psychiatric comorbidity had higher odds of not completing substance treatment relative to those without comorbidity [OR = 1.28 (1.27–1.29)] and an earlier time to attrition [HR = 1.14 (1.13–1.15)]. Psychiatric comorbidity was most strongly associated with treatment non-completion and rate of attrition in those admitted primarily for alcohol [OR = 1.37 (1.34–1.39); HR = 1.19 (1.17–1.21), respectively].

Yet in the population of patients seeking treatment for depression, concurrent SUD does not necessarily predict poor treatment outcome for the mood disorder. In an exploratory analysis of the effect of concurrent substance use disorder on single and combination antidepressants for the treatment of MDD, Davis et al. [20] found no

significant differences between the MDD-SUD and MDD-NSUD groups in terms of dose, time in treatment, response to assigned treatment or remission at Week 12 and 28. Thus, patients with MDD and concurrent SUD are as likely to respond and remit to a single or combination antidepressant treatment as those presenting without SUD. And in a 14-week double-blind trial evaluating the combination of sertraline and naltrexone for the treatment of patients with depression and alcohol dependence, patients in the sertraline-naltrexone group combination were more likely than those in the groups that received only naltrexone, only sertraline, or double placebo to achieve abstinence from alcohol (53.7% vs. 21.3–27.5%) and delay relapse to heavy drinking (98 days vs. 23–29 days) and less likely to be depressed by the end of treatment or report a serious adverse event [74].

These findings highlight the need for the concurrent treatment of depressive symptoms and substance use disorder in patients with MDD-SUD and SIDD. Integrating psychotherapy and pharmacotherapy for comorbid substance use and mood disorders should be considered [10], since reduction in depressive symptoms may improve the course of SUD, and decreases in substance use may contribute to improved mood. Since MDD and SUD are highly co-prevalent, and this comorbidity frequently presents with greater psychopathological and medical severity, as well as worse social function, it is important to treat both MDD and SUD simultaneously, rather than to treat both conditions separately [97].

Although treatment rates for substance use disorders are only half that of major depression, SUD increases the likelihood of depression treatment among comorbid cases [36]. More intensive treatment is often required for MDD patients with concomitant SUD, and an integrated treatment model can permit its implementation. Further controlled trials are needed to examine the impact of each condition upon the other, and to identify optimal treatment approaches for this challenging comorbid presentation.

The only evidence-based guideline offering treatment recommendations for AUD and comor-

bid psychiatric disorders was recently published in Germany, using methodological criteria for the highest quality (“S3-criteria”) as defined by the Association of Scientific Medical Societies in Germany (Preuss et al. 2017). Among these recommendations are that patients with psychosis and comorbid alcohol use disorders should receive psychotherapy after adequate stabilization. Similarly, comorbid depression should be treated beginning 3–4 weeks after withdrawal, and cognitive behavioral therapy has been reported as effective for both depression and AUD, in the majority of studies. For bipolar disorder and AUD, integrated therapy with cognitive behavioral therapy is recommended, with the addition of valproic acid if necessary. An integrated psychotherapeutic treatment strategy is recommended for PTSD patients, with stabilizing, trauma-focused therapeutic interventions. These S3 guidelines based on clinical research are intended to facilitate evidence-based decision-making in psycho- and pharmacotherapy to treat comorbid alcohol use and other psychiatric disorders. As the authors note, further research is urgently needed to support evidence-based clinical approaches to other substance and psychiatric comorbidities.

For the treatment of adults with combined ADHD and SUD, studies have suggested that a combination of pharmacotherapy and psychotherapy is most useful [99, 106]. In an international consensus statement, an expert panel offered recommendations on the treatment of comorbid ADHD and SUD [17]. The authors advised that pharmacotherapy should not be avoided; rather, this approach should be critically encouraged in patients with ADHD and SUD -- with a preference for high doses of long-acting stimulants in ADHD patients with stimulant use disorders, or atomoxetine in patients with alcohol use disorder. Treatment of ADHD can be effective in reducing ADHD symptoms without worsening the substance use disorder. Clinicians should consider treating both ADHD and SUD with their own medication simultaneously, that is, for patients with ADHD and an alcohol use disorder, treatment with atomoxetine and with naltrexone, nalmefene, acamprosate, or

topiramate [17]. This panel recommended a multimodal integrated approach: combining pharmacotherapy (for ADHD and SUD) with a non-pharmacological intervention that targets both the ADHD and SUD, such as an integrated CBT, while noting that further research is needed. Optimizing treatment for comorbid SUD and ADHD can have a significant effect on public health [55]. A recent large registry-based study suggested that individuals with ADHD who received medications had a significant reduction in criminality compared to those who did not, with a relative risk reduction of 32% in men and 41% in women [54].

In sum, the integration of mental health and substance use disorder treatment as well as behavioral treatments such as CBT has demonstrated efficacy [62]. Certain pharmacotherapeutic approaches to the treatment of comorbid psychiatric and substance use disorders have also shown promise. The atypical antipsychotic clozapine has demonstrated preliminary efficacy in the treatment of co-occurring schizophrenia and substance use disorder, as well as naltrexone and disulfiram for comorbid alcohol use disorder, methadone or buprenorphine for opioid use disorder and bupropion and varenicline for tobacco use disorder. Additional research is needed to define optimal pharmacological treatment approaches for these concurrent conditions [1].

Conclusion

Mood, anxiety, and psychotic disorders commonly occur with alcohol and substance use disorders. Exploration of the neurobiology of substance use disorders and mood and anxiety disorders has identified shared neural circuitry in mood, anxiety, and substance use disorders. Genome-wide studies have pointed to, but not fully elucidated, the genetic factors that underlie the high rates of comorbidity between substance use and psychiatric disorders. Environmental factors, such as low parental monitoring and trauma, also play a contributory role in this shared etiology. In light of the extent to which psychiatric and substance use disorders co-occur, careful

clinical consideration should be given to assessing for the presence of one category of disorder in patients presenting for treatment of the other.

Diagnosis of co-occurring disorders requires a comprehensive psychiatric and substance use disorder assessment. Empirically validated screening and diagnostic instruments may be helpful in this regard, but the overlap of symptoms between SUD and psychiatric disorders requires clinical skill to review the history of the emergence of each disorder and their temporal relation to each other.

Treatment of co-occurring disorders should include management of diagnoses simultaneously, often with a combination of pharmacotherapy and psychotherapy, to improve symptom management, and treatment adherence. While pharmacotherapies are often more effective in targeting the non-substance-related comorbidity, available pharmacotherapies which specifically target reducing substance use should be used as available and needed, while integrating pharmacotherapy with evidence-based psychotherapies and behavioral therapies should enhance treatment outcomes for patients with substance and psychiatric comorbidities.

References

1. Akerman SC, Brunette MF, Noordsy DL, Green AI. Pharmacotherapy of co-occurring schizophrenia and substance use disorders. *Curr Addict Rep*. 2014;1(4):251–60.
2. Alegria AA, Hasin DS, Nunes EV, Liu SM, Davies C, Grant BF, Blanco C. Comorbidity of generalized anxiety disorder and substance use disorders: results from the National Epidemiologic Survey on alcohol and related conditions. *J Clin Psychiatry*. 2010;71(9):1187–95. quiz 1252–1253
3. American Psychiatric Association. *Diagnosis and statistical manual of mental disorders (DSM-5)*. Washington DC; 2013.
4. Ameringer KJ, Leventhal AM. Associations between Attention Deficit Hyperactivity Disorder Symptom Domains and DSM-IV Lifetime Substance Dependence. *Am J Addict*. 2013;22(1) <https://doi.org/10.1111/j.1521-0391.2013.00325.x>.
5. Andersen AM, Pietrzak RH, Kranzler HR, Ma L, Zhou H, Liu X, Kramer J, Kuperman S, Edenberg HJ, Nurnberger JI Jr, Rice JP, Tischfield JA, Goate A, Foroud TM, Meyers JL, Porjesz B, Dick

- DM, Hesselbrock V, Boerwinkle E, Southwick SM, Krystal JH, Weissman MM, Levinson DF, Potash JB, Gelernter J, Han S. Polygenic Scores for Major Depressive Disorder and Risk of Alcohol Dependence. *JAMA Psychiat*. 2017;74(11):1153–60.
6. Arranz B, Garriga M, García-Rizo C, San L. Clozapine use in patients with schizophrenia and a comorbid substance use disorder: a systematic review. *Eur Neuropsychopharmacol*. 2018;28(2):227–42.
7. Baigent M. Managing patients with dual diagnosis in psychiatric practice. *Curr Opin Psychiatry*. 2012;25(3):201–5.
8. Bega S, Schaffer A, Goldstein B, Levitt A. Differentiating between bipolar disorder types I and II: results from the National Epidemiologic Survey on alcohol and related conditions (NESARC). *J Affect Disord*. 2012;138:46–53.
9. Blanco C, Alderson D, Ogburn E, et al. Changes in the prevalence of non-medical prescription drug use and drug use disorders in the United States: 1991–1992 and 2001–2002. *Drug Alcohol Dep*. 2007;90:252–60.
10. Blanco C, Alegría AA, Liu SM, Secades-Villa R, Sugaya L, Davies C, Nunes EV. Differences among major depressive disorder with and without co-occurring substance use disorders and substance-induced depressive disorder: results from the National Epidemiologic Survey on Alcohol and Related Conditions. *J Clin Psychiatry*. 2012;73(6):865–73.
11. Boland M, Rielage JK, Hoyt T. The power of negative mood in predicting posttraumatic stress disorder and alcohol abuse comorbidity. *Psychol Trauma*. 2018;10(5):572–5.
12. Breslau J, Kendler KS, Su M, Aguilar-Gaxiola S, Kessler RC. Lifetime risk and persistence of psychiatric disorders across ethnic groups in the United States. *Psychol Med*. 2005;35:317–27.
13. Brown PJ, Wolfe J. Substance abuse and post-traumatic stress disorder comorbidity. *Drug Alcohol Depend*. 1994;35(1):51–9.
14. Chambers RA, Krystal JH, Self DW. A neurobiological basis for substance abuse comorbidity in schizophrenia. *Biol Psychiatry*. 2001;50(2):71–83.
15. Charach A, Yeung E, Climans T, Lillie E. Childhood attention-deficit/hyperactivity disorder and future substance use disorders: comparative meta-analyses. *J Am Acad Child Adolesc Psychiatry*. 2011;50:9–21.
16. Chilcoat HD, Breslau N. Posttraumatic stress disorder and drug disorders: testing causal pathways. *Arch Gen Psychiatry*. 1998;55(10):913–7.
17. Crunelle CL, van den Brink W, Moggi F, Konstenius M, Franck J, Levin FR, van de Glind G, Demetrovics Z, Coetzee C, Luderer M, Schellekens A. ICASA consensus group, Matthys F. International Consensus Statement on Screening, Diagnosis and Treatment of Substance Use Disorder Patients with Comorbid Attention Deficit/Hyperactivity Disorder. *Eur Addict Res*. 2018;24(1):43–51.
18. Crump C, Winkleby MA, Sundquist K, et al. Comorbidities and mortality in persons with schizophrenia: a Swedish national cohort study. A large cohort study demonstrating the premature morbidity and mortality in patients with schizophrenia. *Am J Psychiatry*. 2013;170(3):324–33.
19. Dakwar E, Nunes EV, Bisaga A, Carpenter KC, Mariani JP, Sullivan MA, Raby WN, Levin FR. A comparison of independent depression and substance-induced depression in cannabis-, cocaine-, and opioid-dependent treatment seekers. *Am J Addict*. 2011;20(5):441–6.
20. Davis LL, Pilkinton P, Wisniewski SR, Trivedi MH, Gaynes BN, Howland RH, Zisook S, Balasubramani GK, Fava M, Rush AJ. Effect of concurrent substance use disorder on the effectiveness of single and combination antidepressant medications for the treatment of major depression: an exploratory analysis of a single-blind randomized trial. *Depress Anxiety*. 2012;29(2):111–22.
21. Davis LL, Wisniewski SR, Howland RH, et al. Does comorbid substance use disorder impair recovery from major depression with SSRI treatment? An analysis of the STAR*D level one treatment outcomes. *Drug Alcohol Depend*. 2010;107(2–3):161–70.
22. Donoghue K, Doody GA. Effect of illegal substance use on cognitive function in individuals with psychotic disorder: a review and meta-analysis. *Neuropsychology*. 2012;26(6):785–801.
23. Estévez-Lamorte N, Foster S, Eich-Höchli D, Moggi F, Gmel G, Mohler-Kuo M. Adult attention-deficit/hyperactivity disorder, risky substance use and substance use disorders: a follow-up study among young men. *Eur Arch Psychiatry Clin Neurosci*. 2018 Nov 27; <https://doi.org/10.1007/s00406-018-0958-3>.
24. Fatséas M, Hurmic H, Serre F, Debrabant R, Daulouède JP, Denis C, Auriacombe M. Addiction severity pattern associated with adult and childhood attention deficit hyperactivity disorder (ADHD) in patients with addictions. *Psychiatry Res*. 2016;246:656–62.
25. Foa EB, Huppert JD, Cahill SP. Emotional processing theory: an update. In: Rothbaum BO, editor. *Emotional processing theory: an update*. New York: Emotional Processing Theory; 2006. p. 3–24. An Update.
26. Fowles D. Schizophrenia: diathesis-stress revisited. *Annu Rev Psychol*. 1992;43(1):303–36.
27. Galbraith T, Heimberg RG, Wang S, Schneier FR, Blanco C. Comorbidity of social anxiety disorder and antisocial personality disorder in the National Epidemiological Survey on alcohol and related conditions (NESARC). *J Anxiety Disord*. 2014;28:57–66.
28. Geller B, Cooper TB, Sun K, Zimmerman B, Frazier J, Williams M, Heath J. Double-blind and placebo-controlled study of lithium for adolescent bipolar

- disorders with secondary substance dependency. *J Am Acad Child Adolesc Psychiatry*. 1998;37:171–8.
29. Godette DC, Headen S, Ford CL. Windows of opportunity: fundamental concepts for understanding alcohol-related disparities experienced by young blacks in the United States. *Prev Sci*. 2006;7:311–87.
30. Green AI, Burgess ES, Dawson R, Zimmet SV, Strous RD. Alcohol and cannabis use in schizophrenia: effects of clozapine vs. risperidone. *Schizophr Res*. 2003;60(1):81–5.
31. Green AI, Zimmet SV, Strous RD, et al. Clozapine for comorbid substance use disorder and schizophrenia: do patients with schizophrenia have a reward-deficiency syndrome that can be ameliorated by clozapine? *Harv Rev Psychiatry*. 1999;6(6):287–96.
32. Groenman AP, Greven CU, van Donkelaar MM, Schellekens A, van Hulzen KJ, Rommelse N, Hartman CA, Hoekstra PJ, Luman M, Franke B, Faraone SV, Oosterlaan J, Buitelaar JK. Dopamine and serotonin genetic risk scores predicting substance and nicotine use in attention deficit/hyperactivity disorder. *Addict Biol*. 2016;21(4):915–23.
33. Guo G, Wilhelmsen K, Hamilton N. Gene-lifecourse interaction for alcohol consumption in adolescence and young adulthood: five monoamine genes. *Am J Med Genet B Neuropsychiatr Genet*. 2007;144b:417–23.
34. Gurriarán X, Rodríguez-López J, Flórez G, Pereiro C, Fernández JM, Fariñas E, Estévez V, Arrojo M, Costas J. GenPol study group. Relationships between substance abuse/dependence and psychiatric disorders based on polygenic scores. *Genes Brain Behav*. 2018;4:e12504.
35. Hartz SM, Pato CN, Medeiros H, et al. Comorbidity of severe psychotic disorders with measures of substance use. *JAMA Psychiat*. 2014;71(3):248–54.
36. Hasin D, Kilcoyne B. Comorbidity of psychiatric and substance use disorders in the United States: current issues and findings from the NESARC. *Curr Opin Psychiatry*. 2012;25(3):165–71.
37. Hiemstra M, Nelemans SA, Branje S, van Eijk KR, Hottenga JJ, Vinkers CH, van Lier P, Meeus W, Boks MP. Genetic vulnerability to schizophrenia is associated with cannabis use patterns during adolescence. *Drug Alcohol Depend*. 2018;190:143–50.
38. Hill MN, Gorzalka BB. The endocannabinoid system and the treatment of mood and anxiety disorders. *CNS Neurol Disord Drug Targets*. 2009;8(6):451–8.
39. Hirschfield R, Hasin D, Keller M, Endicott J, Wunder J. Depression and alcoholism: comorbidity in a longitudinal survey. In: Maser J, Cloninger C, editors. *Comorbidity of mood and anxiety disorders*. Washington, DC: American Psychiatric Press; 1990. p. 293–304.
40. Hodgson K, Almasy L, Knowles EE, Kent JW, Curran JE, Dyer TD, Göring HH, Olvera RL, Fox PT, Pearlson GD, Krystal JH, Duggirala R, Blangero J, Glahn DC. Genome-wide significant loci for addiction and anxiety. *Eur Psychiatry*. 2016;36:47–54.
41. Hunt GE, Large MM, Cleary M, Lai HMX, Saunders JB. Prevalence of comorbid substance use in schizophrenia spectrum disorders in community and clinical settings, 1990–2017: systematic review and meta-analysis. *Drug Alcohol Depend*. 2018;191:234–58.
42. Ilbegi S, Groenman AP, Schellekens A, Hartman CA, Hoekstra PJ, Franke B, Faraone SV, Rommelse NNJ, Buitelaar JK. Substance use and nicotine dependence in persistent, remittent, and late-onset ADHD: a 10-year longitudinal study from childhood to young adulthood. *J Neurodev Disord*. 2018;10(1):42.
43. Katzman MA, Bilkey TS, Chokka PR, Fallu A, Klassen LJ. Adult ADHD and comorbid disorders: clinical implications of a dimensional approach. *BMC Psychiatry*. 2017;17(1):302.
44. Kelly TM, Daley DC. Integrated treatment of substance use and psychiatric disorders. *Soc Work Public Health*. 2013;28(3–4):388–406.
45. Kelly TM, Daley DC, Douaihy A. Treatment of substance abusing patients with comorbid psychiatric disorders. *Addict Behav*. 2012;37(1):11–24.
46. Kessler RC. The epidemiology of dual diagnosis. *Biol Psychiatry*. 2004;56:730–7.
47. Kessler RC, Berglund P, Demler O, Jin R, Koretz D, Merikangas KR. National Comorbidity Survey R the epidemiology of major depressive disorder: results from the National Comorbidity Survey Replication (NCS-R). *JAMA*. 2003;289(23):3095–105.
48. Khantzian EJ. The self-medication hypothesis of substance use disorders: a reconsideration and recent applications. *Harv Rev Psychiatry*. 1997;4(5):231–44.
49. Kronenberg LM, Goossens PJ, van Etten DM, van Achterberg T, van den Brink W. Need for care and life satisfaction in adult substance use disorder patients with and without attention deficit hyperactivity disorder (ADHD) or autism spectrum disorder (ASD). *Perspect Psychiatr Care*. 2015;51:4–15.
50. Krawczyk N, Feder KA, Saloner B, Crum RM, Kealhofer M, Mojtabai R. The association of psychiatric comorbidity with treatment completion among clients admitted to substance use treatment programs in a U.S. national sample. *Drug Alcohol Depend*. 2017;1(175):157–63.
51. Larun L, Helseth V, Bramness JG, Haugurud H, Hoie B, Grawe RW, Kornor H, Landheim A, Lossius K, Hammerstrom KT, Waal H. Dual diagnoses – substance use disorder and severe mental illness: Part I – accuracy of screening and diagnostic instruments. Oslo, Norway: Knowledge Centre for the Health Services at The Norwegian Institute of Public Health (NIPH); 2007 Dec. Report from Norwegian Knowledge Centre for the Health Services (NOKK) No. 21–2007.
52. Lee JA, Mavis BE, Stoffelmayr BE. A comparison of problems of life for blacks and whites entering

- substance use programs. *J Psychoactive Drugs*. 1991;23(3):233–9.
53. Levin FR, Evans SM, Vosburg SK, Horton T, Brooks D, Ng J. Impact of attention-deficit hyperactivity disorder and other psychopathology on treatment retention among cocaine abusers in a therapeutic community. *Addict Behav*. 2004;29:1875–82.
 54. Lichtenstein P, Halldner L, Zetterqvist J, Sjolander A, Serlachius E, Fazel S, et al. Medication for attention deficit-hyperactivity disorder and criminality. *N Engl J Med*. 2012;367:2006–14.
 55. Luo S, Levin FR. Towards Precision Addiction Treatment: New Findings in Co-morbid Substance Use and Attention-Deficit Hyperactivity Disorders. *Curr Psychiatry Rep*. 2017;19(3):14.
 56. Lybrand J, Caroff S. Management of schizophrenia with substance use disorders. *Psychiatr Clin N Am*. 2009;32(4):821–33.
 57. McCauley JL, Killeen T, Gros DF, Brady KT, Back SE. Posttraumatic Stress Disorder and Co-Occurring Substance Use Disorders: Advances in Assessment and Treatment. *Clin Psychol (New York)*. 2012;19(3):1.
 58. McFarlane AC. Epidemiological evidence about the relationship between PTSD and alcohol abuse: the nature of the association. *Addict Behav*. 1998;23(6):813–25.
 59. Maremmani AGI, Pani PP, Rovai L, Bacciardi S, Maremmani I. Toward the identification of a specific psychopathology of substance use disorders. *Front Psychiatry*. 2017;27:8:68.
 60. Merikangas KR, Risch NJ, Weissman MM. Comorbidity and co-transmission of alcoholism, anxiety and depression. *Psychol Med*. 1994;24(1):69–80.
 61. Messer T, Lammers G, Müller-Siecheneder F, Schmidt RF, Latifi S. Substance abuse in patients with bipolar disorder: A systematic review and meta-analysis. *Psychiatry Res*. 2017;253:338–50.
 62. Mueser KT, Gingerich S. Treatment of co-occurring psychotic and substance use disorders. *Soc Work Pub Health*. 2013;28(3–4):424–39.
 63. Mueser KT, Rosenberg SD, Xie H, Jankowski MK, Bolton EE, Lu W, et al. Wolfe R. a randomized controlled trial of cognitive-behavioral treatment for posttraumatic stress disorder in severe mental illness. *J Consult Clin Psychol*. 2008;76:259–71.
 64. Ng E, Browne CJ, Samsom JN, Wong AHC. Depression and substance use comorbidity: what we have learned from animal studies. *Am J Drug Alcohol Abuse*. 2017;43(4):456–74.
 65. Nunes EV, Levin FR. Treating depression in substance abusers. *Curr Psychiatry Rep*. 2006;8(5):363–70.
 66. Nunes EV, Levin FR. Treatment of depression in patients with alcohol or other drug dependence: a meta-analysis. *JAMA*. 2004;291(15):1887–96.
 67. Nunes EV, McGrath PJ, Quitkin FM. Treating anxiety in patients with alcoholism. *J Clin Psychiatry*. 1995;56:3–9.
 68. Nunes EV, Rounsaville BJ. Comorbidity of substance use with depression and other mental disorders: from Diagnostic and Statistical Manual of Mental Disorders, fourth edition (DSM-IV) to DSM-V. *Addiction*. 2006;101(Suppl. 1):89–96.
 69. Ojeda VD, McGuire TG. Gender and racial/ethnic differences in use of outpatient mental health and substance use services by depressed adults. *Psychiatry Q*. 2006;77(3):211–22.
 70. Oliveira LM, Bermudez MB, Macedo MJA, Passos IC. Comorbid social anxiety disorder in patients with alcohol use disorder: a systematic review. *J Psychiatr Res*. 2018;106:8–14.
 71. Østergaard MLD, Nordentoft M, Hjorthøj C. Associations between substance use disorders and suicide or suicide attempts in people with mental illness: a Danish nation-wide, prospective, register-based study of patients diagnosed with schizophrenia, bipolar disorder, unipolar depression or personality disorder. *Addiction*. 2017;112(7):1250–9.
 72. Owen RR, Fischer EP, Booth BM, et al. Medication noncompliance and substance abuse among patients with schizophrenia. *Psychiatr Serv*. 1996;47(8):853–8.
 73. Parmar A, Kaloiya G. Comorbidity of personality disorder among substance use disorder patients: A narrative review. *Indian J Psychol Med*. 2018;40(6):517–27.
 74. Pettinati HM, Oslin DW, Kampman KM, et al. A double-blind, placebo-controlled trial combining sertraline and naltrexone for treating co-occurring depression and alcohol dependence. *Am J Psychiatry*. 2010;167(6):668–75.
 75. Polimanti R, Agrawal A, Gelernter J. Schizophrenia and substance use comorbidity: a genome-wide perspective. *Genome Med*. 2017;9(1):25.
 76. Preuss UW (2008) Abhängigkeitserkrankungen und Psychosen: Erkenntnisse und Kontroversen in Vorbereitung auf neue Klassifikationssysteme. *Die Psychiatrie* 2:61–68. Cited in Preuss et al. 2018.
 77. Preuss UW, Gouzoulis-Mayfrank E, Havemann-Reinecke U, Schafer I, Beutel M, Hoch E, Mann KF. Psychiatric comorbidity in alcohol use disorders: results from the German S3 guidelines. *Eur Arch Psychiatry Clin Neurosci*. 2018;268(3):219–29.
 78. Prisciandaro JJ, Brown DG, Brady KT, Tolliver BK. Comorbid anxiety disorders and baseline medication regimens predict clinical outcomes in individuals with co-occurring bipolar disorder and alcohol dependence: Results of a randomized controlled trial. *Psychiatry Res*. 2011;188(3):361–5.
 79. Quednow BB, Brinkmeyer J, Mobascher A, Nothnagel M, Musso F, Grunder G, et al. Schizophrenia risk polymorphisms in the TCF4 gene interact with smoking in the modulation of auditory sensory gating. *Proc Natl Acad Sci U S A*. 2012;109:6271–6.
 80. Redmond ML, Watkins DC, Broman CL, Abelson JM, Neighbors HW. Ethnic and gender differences

- in help seeking for substance disorders among black Americans. *J Racial Ethn Health Disparities*. 2017;4(2):308–16.
81. Regier DA, Farmer ME, Rae DS. Comorbidity of mental disorders with alcohol and other drug abuse. Results from the Epidemiologic Catchment Area (ECA) Study *JAMA*. 1990a;264:2511–8.
82. Roberts NP, Roberts PA, Jones N, Bisson JI. Psychological therapies for post-traumatic stress disorder and comorbid substance use disorder. *Cochrane Database Syst Rev*. 2016a;4:CD010204.
83. Regier DA, Farmer ME, Rae DS, Locke BZ, Keith SJ, Judd LL, Goodwin FK. Comorbidity of mental disorders with alcohol and other drug abuse. Results from the epidemiologic catchment area (ECA) study. *JAMA*. 1990b;264(19):2511–8.
84. Roberts NP, Roberts PA, Jones N, Bisson JI. Psychological therapies for post-traumatic stress disorder and comorbid substance use disorder. *Cochrane Database Syst Rev*. 2016b;4(4):CD010204.
85. Roth RM, Brunette MF, Green AI. Treatment of substance use disorders in schizophrenia: a unifying neurobiological mechanism? *Curr Psychiatry Rep*. 2005;7(4):283–91.
86. San L, Arranz B, Martínez-Raga J. Antipsychotic drug treatment of schizophrenic patients with substance abuse disorders. *Eur Addict Res*. 2007;13:230–43.
87. Sanchez K, Walker R, Campbell AN, Greer TL, Hu MC, Grannemann BD, Nunes EV, Trivedi MH. Depressive Symptoms and Associated Clinical Characteristics in Outpatients Seeking Community-Based Treatment for Alcohol and Drug Problems. *Subst Abus*. 2015;36(3):297–303.
88. Schellekens AF, de Jong CA, Buitelaar JK, Verkes RJ. Co-morbid anxiety disorders predict early relapse after inpatient alcohol treatment. *Eur Psychiatry*. 2015;30(1):128–36.
89. Scott JG, Matuschka L, Niemelä S, Miettunen J, Emmerson B, Mustonen A. Evidence of a causal relationship between smoking tobacco and schizophrenia spectrum disorders. *Front Psych*. 2018;9:607. <https://doi.org/10.3389/fpsy.2018.00607>.
90. Sher L, Stanley BH, Harkavy-Friedman JM, et al. Depressed patients with co-occurring alcohol use disorders: a unique patient population. *J Clin Psychiatry*. 2008;69(6):907–15.
91. Shiner B, Leonard Westgate C, Bernardy NC, Schnurr PP, Watts BV. Trends in Opioid Use Disorder Diagnoses and Medication Treatment Among Veterans With Posttraumatic Stress Disorder. *J Dual Diagn*. 2017;13(3):201–12.
92. Siris SG. Pharmacological treatment of substance-abusing schizophrenic patients. *Schizophr Bull*. 1990;16(1):111–22.
93. Soler Artigas M, Sánchez-Mora C, Rovira P, Richarte V, García-Martínez I, Pagerols M, Demontis D, Stringer S; ADHD Group of the Psychiatric Genomics Consortium, International Cannabis Consortium, Vink JM, Børghlum AD, Neale BM, Franke BI, Faraone SV, Casas M, Ramos-Quiroga JA, Ribasés M. Attention-deficit/hyperactivity disorder and lifetime cannabis use: genetic overlap and causality. *Mol Psychiatry*. 2019; <https://doi.org/10.1038/s41380-018-0339-3>.
94. Swartz MS, Wagner HR, Swanson JW, et al. Substance use in persons with schizophrenia: baseline prevalence and correlates from the NIMH CATIE study. *J Nerv Ment Dis*. 2006;194(3):164–72.
95. Szaflarski M, Bauldry S, Cubbins LA, Meganathan K. Nativity, Race-Ethnicity, and Dual Diagnosis among US Adults. *Res Sociol Health Care*. 2017;35:171–91.
96. Tamburin S, Federico A, Morbioli L, Faccini M, Casari R, Zamboni L, Briguglio G, Lugoboni F. Screening for adult attention deficit/hyperactivity disorder in high-dose benzodiazepine dependent patients. *Am J Addict*. 2017;26(6):610–4.
97. Tirado Muñoz J, Farré A, Mestre-Pintó J, Szerman N, Torrens M. Dual diagnosis in Depression: treatment recommendations. *Adicciones*. 2018;30(1):66–76.
98. Torrens M, Martín-Santos R, Samet S. Importance of clinical diagnoses for comorbidity studies in substance use disorders. *Neurotox Res*. 2006;10(3–4):253–61.
99. Torrens M, Rossi PC, Martínez-Riera R, Martínez-Sanvisens D, Bulbena A. Psychiatric co-morbidity and substance use disorders: treatment in parallel systems or in one integrated system? *Subst Use Misuse*. 2012;47(8–9):1005–14.
100. Trudeau KJ, Burtner J, Villapiano AJ, Jones M, Butler SF, Joshi K. Burden of schizophrenia or psychosis-related symptoms in adults undergoing substance abuse evaluation. *J Nerv Ment Dis*. 2018;206(7):528–36.
101. van Emmerik-van Oortmerssen K, Vedel E, Kramer FJ, Koeter MW, Schoevers RA, van den Brink W. Diagnosing ADHD during active substance use: feasible or flawed? *Drug Alcohol Depend*. 2017;180:371–5.
102. van Emmerik-van Oortmerssen K, van de Glind G, van den Brink W, Smit F, Crunelle CL, Swets M, Schoevers RA. Prevalence of attention-deficit hyperactivity disorder in substance use disorder patients: a meta-analysis and meta-regression analysis. *Drug Alcohol Depend*. 2012;122:11–9.
103. Varo C, Murru A, Salagre E, Jiménez E, Solé B, Montejó L, Carvalho AF, Stubbs B, Grande I, Martínez-Arán A, Vieta E, Reinares M. Behavioral addictions in bipolar disorders: a systematic review. *Eur Neuropsychopharmacol*. 2018;S0924-977X(18)30847–2.
104. Volkow ND. The reality of comorbidity: depression and drug abuse. *Biol Psychiatry*. 2004;56(10):714–7.
105. Walker EF, Diforio D. Schizophrenia: a neural diathesis-stress model. *Psychol Rev*. 1997;104(4):667.
106. Zulauf CA, Sprich SE, Safren SA, Wilens TE. The complicated relationship between attention deficit/hyperactivity disorder and substance use disorders. *Curr Psychiatry Rep*. 2014;16:436.